

Global CO₂ Emissions Tracker by Sector

1. Introduction:

This project focuses on analysing global carbon dioxide (CO₂) emissions by country and economic sector over time. The goal is to develop an interactive Power BI dashboard that enables users to track emission trends, identify top emitting sectors and countries, and derive insights for environmental policy and awareness.

2. Abstract:

Using a long-format dataset sourced from the CORGIS emissions dataset, this analysis highlights trends in CO₂ emissions across sectors such as Power, Transport, Buildings, Other Industry, and Other Sectors. The dashboard offers a country-wise and sector-wise breakdown from 1970 to 2012, empowering users to explore historical emission trends, compare country performance, and understand emissions per capita and per GDP.

3. Tools Used:

- Microsoft Power BI
- Python (for data cleaning and reshaping with Pandas)
- Excel (for initial review and formatting)

4. Steps Involved in Building the Project:

- **Data Preparation:** Cleaned and standardized wide-format emissions data using Python. Renamed sector columns for readability and reshaped it to long format using `melt()` for easier visualization.
- **Dashboard Development:** Built multiple visuals including:
 - Stacked Area Chart showing emissions by sector over time
 - Map visual for country-wise total and per capita emissions
 - Bar chart comparing top 10 emitting countries
 - Slicers for dynamic filtering by Year and Sector
 - KPI Cards to highlight Total Emissions and Top Emitting Country
- **Interactivity & Design:** Enabled sector and year filtering; added clear titles, consistent colour themes, and visual alignment for clarity.

5. Conclusion:

The dashboard provides a clear and interactive overview of global CO₂ emissions patterns. It highlights the dominance of power and transport sectors in total emissions and shows which countries have contributed the most over time. By enabling comparisons across regions and years, this tool can support educational, analytical, and policy-driven use cases focused on climate change and sustainability.